Lexical Task Heads on Simple Tasks

The best approaches so far, compared

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Setting and how to read the numbers

Llama-3.1-8B-Instruct (bf16), the 14 simple tasks of Todd et al. (antonym, capitals, translations, ...), rule style unless stated. A head set is tested by patching its mean output ($\times\alpha$) at the last prompt token:

- **Check 1**: zero-shot prompts, which do not say what the task is (unpatched accuracy ≈ 0). The main test: does the set *bring in* the task?
- **Check 2**: 1-shot prompts the model answered wrongly (baseline 0). The prompt already shows the task, so this check tests repair, not task transfer.

Readouts: **Logit-head** (head write decoded by the unembedding, as Yang et al.) and **J-head** (Head-JLens: head output through its fitted Jacobian, then decoded). Accuracy is the flexible score. Unless stated, a number is the **mean over the 14 tasks in check 1**.

The numbers come from three protocols, marked in Table 1: **held-out** (E4: heads, means and α chosen on one half of the items, scored on the other; the fairest), **same items** (chosen and scored on the same items, best α: optimistic), and **one fold** (held-out items, but best α in fold 1 only). Every run is single, greedy, without intervals.

1 Starting point A: lexical task heads (Yang et al., 2026)

Their method. A head is a lexical task head if, on $\geq$10% of 100 correctly answered prompts, one of its 10 top decoded tokens (logit lens) is in a hand-written term list for the task (2–22 tokens, e.g. *opposite*, *capital*). The set's mean output is then patched into prompts without task information.

What we changed.

1. *Padding masked.* Their code batches prompts padded on the left without an attention mask, so the model attends to the padding. We mask it: each prompt is computed as if run alone.
2. *J-head* as a second readout, beside their Logit-head.
3. *Scoring.* Their first-token match marks any number correct and capitalised German nouns wrong; we generate the reply and use a flexible score (whole answer as words, case/accents/articles ignored). Heads, means and patched items are all chosen by it.
4. *Second check.* The paper's zero-shot figures (App. G.1) are reproduced only by the setting of their one released script: 1-shot prompts answered wrongly. We report both checks.
5. bf16 instead of fp32 (the precision the lens was fitted in).

How it worked.

- $\sim$ *Replication.* Head counts within one head of Yang's code-setting figures. Check 1 works only on fact-like tasks (capitals, landmark, park, product); nothing on antonym, synonym, present–past, prev-item and translations. Check 2 gains on all 14 and matches their figures.
- ✓ *J-head.* Finds more heads (on 23/28 task $\times$ style pairs), the extra ones in layers 10–19; the heads only J-head finds have effects of their own; its lead holds at equal set size (16/25 leads at 5, 10 and 20 heads, none reversed). It does not contain the Logit-head set (H2 falsified).
- ✗ *Which templates.* Heads that pass under all 5 templates help only fact tasks; restricting the union to heads shared across tasks removes its gains.
- **Held-out:** the E1 J-head set reaches **0.20**. It does nothing on antonym, singular–plural, present–past and the three translations ($\leq$0.03).

2 Starting point B: instruction function vector (Davidson et al., 2025)

Their method. Each head's causal score: replace the prompt's instruction by uninformative text (equiprobable tokens, WikiText, other tasks' instructions; matched in length and log-probability), set the head to its mean output, and measure the gain in the answer's probability. The 20 heads with the highest score averaged over tasks form one set; the function vector (FV) is the sum of their mean outputs, added once after layer $|L|/3$ (layer 11) at the last token, $\alpha = 1$.

What we changed.

1. *Instructions and scoring.* Our rule templates inst1–4 as the instructions (25 correct prompts each for the mean), flexible score.
2. *Test.* Our check 1 with its control and an α grid, instead of their evaluation.
3. *Set size and scope.* 4 tasks first, then heads chosen over all 14 tasks (30 heads); a 5–1,024 size sweep.
4. *Per-task ranking, held out* (E4): the first N heads by each task's own score, computed on the choosing half: FV-20, FV-40, FV-N_t (as many heads as CC-pooled, below).
5. *Generic heads removed* (FV split): 6 of the 30 heads decode the same tokens on every task (line ends, colons), attend to the answer slot, and give one constant output. FV minus these 6 = *task24*.

How it worked.

- ✓ *Their 30-head FV* at $\alpha = 1$: **0.39** on held-out items, twice E1 (0.20); in its first run (same items) $\geq$ our best E1 set on 11/14 tasks. Nothing at $|\alpha| \geq 3$.
- ✓ *Per-task ranking*: FV-N_t **0.59**, the best held-out set; also best at 20 (0.38) and 40 heads (0.43). It alone carries present–past (0.85). It fails on English–German (0.01) and is below CC-pooled on all three translations and prev-item.
- ✓ *Generic heads removed*: task24 **0.58** vs the full FV 0.33 on the same items, $\geq$ it on all 14 tasks; translations 0.39–0.70 (FV $\leq$0.10). The 6 generic heads are most of the FV's norm, in one direction shared by all tasks; scaled up alone, they turn replies into *://* or *1. 2. 3.*; without them the FV works best at α 2–3 instead of 1. Removing 6 random heads does not help.
- ∼ *Reading the FV heads*: under J-head about 20 of 30 are readable (answer category: L16H25 *cities*, L17H4; domain; relation: L13H22 *opposite, next, earlier*). Only 3–11 per task are in our decoded-word sets. A lens-only estimate of their score finds 7–8 of the 30 (late layers), not the mid-layer word heads.

3 What we built on them

CC-pooled: term lists from decoded words (Yang's criterion, our lists; J-head). Instead of the hand-written list, 5 *task words* (decoded on this task more than on the other 13) + 5 *common words* (decoded on every task: *what, which, maybe, https*), with words and heads counted on the 5 templates pooled. 95–154 heads. ✓ Held-out **0.44** vs E1 0.20 and layer-matched random sets 0.02; above E1 on 13/14. Carries the tasks E1 cannot (antonym, singular–plural, translations 0.38–0.40). Fails on present–past (0.02). At 20 and 40 heads it falls to 0.22 / 0.26, below FV at the same size. Both word kinds and the pooling are needed; removing WikiText-generic words helps facts and translations but hurts antonym, synonym and sequences.

Attend + decode (Yang's readout, ranking by mechanism). Heads that attend to the instruction's task words and decode a task word, after L16H25 (attends *capital*, *country*; decodes *city*). ∼ Few heads carry named answer types: capitals 0.88 with L16H25 alone, landmark 0.79 (5 heads), product–company 0.72–0.80 (20 heads). Nothing on translations, sequences, present–past; mean 0.25 at 20 heads (0.28, one fold).

LLM-scored heads (E5; GPT-5.5 reads each head's top-20 decoded tokens and scores it 0–100 for representing the *task*, not the answer or the input). Run on 2 tasks only, held-out. ∼ Country–capital: J-head 20 heads 0.78, Logit-head 0.70, ≈ CC-20 0.77, below FV-20 0.88. English–German: nothing in check 1 below 60 heads, then J-head 0.21–0.26 vs full CC 0.39; Logit-head sets 0 at every size. J-head sets beat Logit-head sets on English–German at every size and on country–capital at 20–30 heads (not at 40: 0.59 vs 0.68).

Single heads that carry a task. L16H25 for capitals (above). L15H1 for present–past: with example-style means it alone gives 0.80 (check 1) and 0.76 (check 2), control 0, while the 4-head threshold set gives 0; J-head reads past participles from it, Logit-head reads nothing.

4 Side by side

Head set	Origin	Heads	Check 1	Check 2	Protocol
E1, J-head	Yang + J-head	7–50	0.20	0.31	held-out
CC-pooled	Yang, our lists	97–154	0.44	0.33	held-out
CC-20 / CC-40	Yang, our lists	20 / 40	0.22 / 0.26	0.28 / 0.34	held-out
FV-30 ($\alpha = 1$)	Davidson, as published	30	0.39	–	held-out
FV-20 / FV-40	Davidson, per task	20 / 40	0.38 / 0.43	0.33 / 0.39	held-out
FV-N_t	Davidson, per task	97–154	**0.59**	0.39	held-out
Random-N_t	control	97–154	0.02	0.12	held-out
FV-univ-20	Davidson, over 14 tasks	20	0.63	0.39	one fold
Attend + decode-20	Yang, by mechanism	20	0.28	0.35	one fold
task24	Davidson minus generic	24	**0.58**	–	same items
FV-30, same run	Davidson, as published	30	0.33	–	same items

Table 1: Mean accuracy over the 14 tasks. Held-out: cross-fitted α, mean of the two folds. One fold: best α in fold 1. Same items: best α on the items scored (task24 at a fixed $\alpha = 3$: 0.54). Rows of different protocols are not directly comparable. –: not run in check 2. LLM-scored sets (2 tasks only) are not in the table.

By task type (check 1; per task in App. A):

- *Fact tasks* (capitals, landmark, park, product): every method works; FV-N_t 0.88–0.94; a single head can be enough (L16H25).
- *Translations and prev-item*: among the held-out sets, only CC-pooled (0.38–0.40 vs FV-N_t 0.01–0.25). FV heads do carry them when ranked over all 14 tasks (FV-univ-20, one fold: 0.32–0.55) or without the generic heads (task24, same items: 0.39–0.70).
- *Present–past*: only FV heads ranked by their score (FV-N_t 0.85; FV-univ-20 0.80, one fold), or L15H1 with example means. Not the 30-head FV or task24 ($\leq$0.05), and no decoded-word set with rule means.
- *Antonym, synonym, singular–plural, next-item*: FV-N_t first (0.49–0.85), CC-pooled second (0.26–0.69), E1 near 0.

5 What did not help

- ✗ Heads passing under all 5 templates ($k = 5$), and strict consensus lists: better on fact tasks only.
- ✗ Union restricted to heads shared across tasks: loses the union's gains.
- ✗ Heads from what the instruction adds (E6, clean − corrupted readings): 0.18 vs CC 0.44; the heads whose words do not depend on the instruction are needed.
- ✗ Word lists without the cross-task contrast (task minus WikiText, 0.21) or with a within-group contrast (0.32).

6 Main insights

1. **J-head is the better readout** for finding task heads: more heads, earlier layers, causal effects that remain at equal size; LLM-scored J-head sets also mostly beat Logit-head ones.
2. **The hand-written term list is the bottleneck of Yang's method**, not the criterion: lists built from what heads actually decode double the held-out effect (0.20 → 0.44).
3. **A causal ranking still finds better heads** than any decoding rule (FV-N_t 0.59). Per-task FV rankings miss translations and prev-item, where CC-pooled wins; FV heads ranked over all 14 tasks do not (FV-univ-20, 0.63, one fold only).
4. **Davidson's FV is held back by a few generic heads** that add a large, task-independent direction; without them it works at larger α and also carries translations.
5. **Check 2 does not separate the sets** (all sets 0.28–0.39, random sets 0.12, control means up to 0.13): the prompt already gives the task. Check 1 is the informative test.
6. **Read the replies.** 19–33% of patched replies on several tasks contain the answer after a prefix the score rejects; part of a "loss" can be a format change.

Open: task24 held-out (only same-items so far); is the generic FV part harmful through its heads or only its norm; LLM-scored heads on the other 12 tasks; E4b (new wordings, held-out wordings) not run.

A Per-task results

Tables 2–6 give every task. They are built by `scripts/best_approaches_tables.py` from the existing result files (E4 tables, FV split report, E5 tables); nothing is recomputed. In Tables 2, 4 and 5 the largest value per task is in bold within the held-out main sets (or within all30/task24); the other columns are measured differently and are not compared in bold.

	held-out					one fold			same items	
Task	E1	CC	FV-30	FV-N_t	Random	CC-clean	A+D-20	FV-univ-20	FV-30	task24
antonym	0.03	0.42	0.25	**0.49**	0.00	0.17	0.14	0.55	0.26	0.61
synonym	0.11	0.34	0.04	**0.55**	0.02	0.08	0.11	0.24	0.04	0.35
singular–plural	0.01	0.69	0.68	**0.85**	0.00	0.61	0.65	0.90	0.59	0.92
present–past	0.00	0.02	0.05	**0.85**	0.00	0.03	0.05	0.80	0.02	0.04
country–capital	0.58	0.75	0.76	**0.94**	0.02	0.77	0.74	0.98	0.83	1.00
country–currency	0.12	0.29	0.36	**0.47**	0.00	0.39	0.23	0.72	0.13	0.64
landmark–country	0.68	0.92	0.89	**0.93**	0.04	0.95	0.60	0.95	0.89	0.95
park–country	0.44	0.73	0.87	**0.88**	0.02	0.67	0.28	0.88	0.89	0.91
product–company	0.48	0.35	0.79	**0.91**	0.15	0.68	0.80	0.86	0.63	0.82
next-item	0.22	0.26	0.19	**0.70**	0.00	0.09	0.05	0.31	0.11	0.30
prev-item	0.05	**0.26**	0.04	0.19	0.00	0.18	0.06	0.34	0.01	0.04
English–French	0.01	**0.40**	0.15	0.21	0.00	0.37	0.02	0.43	0.09	0.50
English–German	0.02	**0.39**	0.08	0.01	0.00	0.37	0.11	0.32	0.02	0.39
English–Spanish	0.00	**0.38**	0.24	0.25	0.01	0.17	0.02	0.55	0.10	0.70
Mean	0.20	0.44	0.39	0.59	0.02	0.39	0.28	0.63	0.33	0.58
Control mean	0.03	0.06		0.04						

Table 2: Check 1 (zero-shot prompts), every approach, per task. *Held-out*: E4, cross-fitted α, mean of the two folds; E1 and CC J-head sets rechosen on the choosing half; FV-30 = Davidson et al.'s 30-head function vector at $\alpha = 1$; FV-N_t = as many heads as CC, ranked by the task's own FV score; Random = 5 layer-matched draws of CC's size at CC's α. *One fold*: E4 secondary sets, fold 1, best α. *Same items*: FV split run (FV-30: the full FV; task24: FV-30 minus its 6 generic heads), best of $\alpha \in \{0, 1, 2, 3, 5, 8\}$ on the items scored. Control mean: the same heads patched with another task's mean (held-out sets only).

Task	CC-20	FV-20	Random-20	CC-40	FV-40	Random-40
antonym	0.01	0.40	0.02	0.09	0.38	0.00
synonym	0.10	0.13	0.03	0.19	0.28	0.02
singular–plural	0.25	0.42	0.01	0.27	0.62	0.01
present–past	0.03	0.87	0.01	0.04	0.68	0.00
country–capital	0.77	0.88	0.01	0.74	0.92	0.00
country–currency	0.14	0.14	0.00	0.14	0.11	0.00
landmark–country	0.69	0.94	0.01	0.95	0.95	0.02
park–country	0.51	0.87	0.01	0.60	0.89	0.02
product–company	0.51	0.66	0.18	0.40	0.77	0.24
next-item	0.03	0.02	0.01	0.16	0.26	0.02
prev-item	0.03	0.04	0.00	0.07	0.03	0.00
English–French	0.00	0.00	0.00	0.01	0.03	0.01
English–German	0.00	0.00	0.00	0.01	0.00	0.00
English–Spanish	0.00	0.00	0.00	0.00	0.05	0.01
Mean	0.22	0.38	0.02	0.26	0.43	0.02

Table 3: Check 1 at equal size (held-out, cross-fitted α). CC-N: the first N heads by CC match rate; FV-N: the first N by the task's FV score; Random-N: layer-matched to CC-N.

		held-out				one fold		control mean		
Task	items	E1	CC	FV-N_t	Random	A+D-20	FV-univ-20	E1	CC	FV-N_t
antonym	100/100	0.32	**0.35**	**0.35**	0.03	0.34	0.31	0.02	0.07	0.08
synonym	100/100	0.15	0.17	**0.18**	0.05	0.13	0.15	0.07	0.10	0.09
singular–plural	11/15	0.69	**0.77**	0.65	0.08	0.71	0.69	0.13	0.47	0.25
present–past	30/31	0.41	0.64	**0.82**	0.28	0.72	0.88	0.10	0.13	0.07
country–capital	14/13	0.76	**0.77**	0.74	0.17	0.66	0.67	0.15	0.11	0.07
country–currency	31/34	0.14	**0.23**	0.17	0.21	0.24	0.32	0.00	0.00	0.06
landmark–country	100/87	**0.34**	0.32	0.32	0.13	0.36	0.34	0.14	0.14	0.24
park–country	22/23	0.21	0.33	**0.38**	0.20	0.33	0.28	0.22	0.07	0.09
product–company	84/75	0.21	0.16	**0.22**	0.10	0.17	0.17	0.01	0.00	0.01
next-item	54/50	0.42	0.26	**0.61**	0.04	0.50	0.59	0.61	0.24	0.32
prev-item	59/63	0.11	0.09	**0.32**	0.07	0.08	0.22	0.16	0.34	0.29
English–French	100/100	0.21	0.16	**0.36**	0.09	0.15	0.23	0.01	0.04	0.03
English–German	100/100	0.20	**0.21**	0.17	0.10	0.29	0.32	0.01	0.01	0.01
English–Spanish	100/100	0.18	**0.24**	0.22	0.14	0.25	0.33	0.08	0.07	0.07
Mean		0.31	0.33	0.39	0.12	0.35	0.39	0.12	0.13	0.12

Table 4: Check 2 (1-shot prompts answered wrongly, baseline 0), per task. Items: patched items in fold 1 / fold 2 (few on several tasks). Held-out and one-fold sets as in Table 2. Control means are often as large as the task means (next-item, prev-item, singular–plural).

Task	all30	task24	generic6	drop-6 random	α all30	α task24
antonym	0.26	**0.61**	0.04	0.20	1	3
synonym	0.04	**0.35**	0.04	0.04	0	3
singular–plural	0.59	**0.92**	0.03	0.59	1	2
present–past	0.02	**0.04**	0.00	0.02	1	3
country–capital	0.83	**1.00**	0.02	0.48	1	2
country–currency	0.13	**0.64**	0.00	0.13	1	3
landmark–country	0.89	**0.95**	0.01	0.67	1	2
park–country	0.89	**0.91**	0.10	0.83	1	2
product–company	0.63	**0.82**	0.11	0.61	1	2
next-item	0.11	**0.30**	0.08	0.12	2	8
prev-item	0.01	**0.04**	0.01	0.01	1	8
English–French	0.09	**0.50**	0.05	0.08	1	3
English–German	0.02	**0.39**	0.00	0.01	1	3
English–Spanish	0.10	**0.70**	0.01	0.04	1	3
Mean	0.33	0.58	0.04	0.27		

Table 5: Davidson et al.'s 30-head FV split (check 1, same items, best α per task over $\{0, 1, 2, 3, 5, 8\}$). all30: the full FV; task24: without the 6 generic heads; generic6: those 6 alone; drop-6 random: the FV minus 6 random non-generic heads, mean of 3 draws. Last two columns: the α of the best value.

		check 1					check 2		
Task	Heads	LLM-J	LLM-L	CC	FV	Random	LLM-J	CC	FV
country–capital	20	0.78	0.70	0.77	0.88	0.01	0.78	0.76	0.67
	30	0.76	0.68	0.74	0.91	0.01	0.74	0.77	0.73
	40	0.59	0.68	0.74	0.92	0.05	0.79	0.84	0.75
	60	0.49	–	0.61	0.94	0.01	0.71	0.73	0.78
	80	0.59	–	0.64	0.95	0.04	0.70	0.83	0.77
E1 set (34–35 heads): check 1 0.58, check 2 0.76									
English–German	20	0.00	0.00	0.00	0.00	0.00	0.19	0.13	0.16
	30	0.00	0.00	0.00	0.00	0.00	0.25	0.16	0.24
	40	0.01	0.00	0.01	0.00	0.00	0.23	0.18	0.24
	60	0.21	–	0.05	0.00	0.00	0.28	0.20	0.21
	80	0.26	–	0.30	0.00	0.00	0.28	0.22	0.18
	97–99	0.25	–	0.39	0.01	0.00	0.28	0.21	0.17
E1 set (19 heads): check 1 0.02, check 2 0.20									

Table 6: E5, LLM-scored heads (GPT-5.5 scores every head from its top-20 decoded tokens), the two tasks run so far; held-out, cross-fitted α, mean of the two folds. LLM-J / LLM-L: top-N heads by score on J-head / Logit-head readings (Logit-head sets run up to 40 heads). CC, FV: E4 sets of the same size (last English–German row: the full CC set and FV-N_t). Random: layer-matched to LLM-J. –: not run.